

Nikolas Dinh

909-636-7988 | nikolasdinh5@gmail.com | <https://www.linkedin.com/in/nikolas-dinh/>

EDUCATION

California Polytechnic University, Pomona

Expected graduation class of 2027

Mechanical Engineering Student 3.5 GPA

- Coursework: Properties of Materials, Mechanics of Materials, Lin Algebra and Differential Equations, Physics Series (Mechanics, E&M, Modern Physics), Chemistry, MATLAB, Dynamics, Autocad, Fusion, Arduino IDE, Electrical Engineering, Python

SKILLS

Technical Skills: SolidWorks (**CSWP**), Python, Instron, Onshape, Autocad, MATLAB, Fusion 360, 3D printing, Welding (MIG, TIG, Stick, Flux Core, Oxy-Fuel), ANSYS Fluent, GD&T

PROJECTS

Sensor-Based Modeling and Interfaces – MATLAB & Raspberry Pi Projects

January 2025 - February 2025

- Developed a MATLAB system integrating Raspberry Pi SenseHAT gyroscope and joystick to rotate a displayed image based on device orientation or user control, optimizing for real-time responsiveness and interface clarity.
- Modeled air conditioning performance using MATLAB by analyzing temperature, lux, and insulation data collected with Vernier probes; predicted future thermal behavior in residential settings using custom data analysis scripts.
- Iterated through multiple versions of code to streamline logic, improve readability, and support future scaling for both UI systems and predictive models.

ANSYS Beam Wing FSAE Intern Project

October 2025 - December 2025

- Researched and selected NACA airfoil geometries to design a rear beam wing for Cal Poly Pomona's Formula SAE vehicle maintaining a 6 week research and development timeline
- Conducted 21+ CFD simulations with SolidWorks and in ANSYS Fluent to evaluate downforce, wake behavior, and aerodynamic efficiency with drag considering other aerodynamic tools
- Authored an 18-page technical report documenting methodology, simulation results, errors, and design trade-offs.
- Presented findings to 50+ team members to assess cost-benefit and system-level aero integration.
- Achieved ~5 lb increase in downforce and rearward center-of-pressure shift; identified wake interference as a limiting factor.

Candy Crushing Device – 3D Design, Testing, and Assembly

March 2025 - June 2025

- Designed a handheld candy grinding device using burr-mill inspired geometry, applying iterative prototyping and CAD modeling techniques in Fusion 360.
- Modeled 15+ components including burrs, dispenser tabs, spacers, and acrylic panels; produced full assemblies and exploded views for design verification and documentation with proper ANSI standard dimensioning.
- Integrated revisions in sliding tab mechanisms and a drill-driven burr system to control candy feed rate and crushing, with final design capable of dispensing ≤ 9 mm fragments as required.
- Collaborated in a two-person team, managing deadlines, interpersonal skills, promoting reliability and efficiency.
- Authored a formal design report with full design rationale, operation manual, exploded-view drawings, and future design

Knife Design and Fabrication Project

March 2025 - June 2025

- Engineered a functional fixed-blade knife using A2 tool steel, applying principles of material science and heat treatment to achieve a post-temper hardness of 51.5 HRC and maximum bending load of 13.65 KN using Universal Instron
- Modeled after a Japanese-style Gyuto knife with a high saber grind and custom PLA Wa-style handle with friction fit sheath using SolidWorks and 3D printing techniques, to prioritize functionality in both chopping and slicing.
- Conducted full fabrication cycle including beveling, tempering, and sharpening using hand tools and a wet stone from McMaster-Carr stock material, achieving the fastest slicing time (0.79 s) in the final evaluation.
- Collaborated in a four-person team to manage material selection, thermomechanical processing, and testing methodology, integrating results into a 30+ page formal report and concise 5 minute presentation of processes.

WORK EXPERIENCE

Auto Mechanic Assistant

June 2019-Present

Team Automotive Buena Park

Buena Park, CA

- Assisted in routine vehicle check ups regarding fluids and wear based products
- Collaborated with senior mechanics replacing differentials, engines, drive trains, and chassis
- Gained in shop experience developing hard skills in using various power tools and diagnostic machines for auto part analysis